

VR-based Vector Watercolor Painting System

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ABSTRACT

We develop a VR-based watercolor painting system. The system generates paintings in the form of vector graphics, which remain clear under free scaling and panoramic projection in VR space. Meanwhile, in order to create a better operation experience and more convenient user interaction in the VR environment, we optimize the system to enable a variety of realistic watercolor effects and provide users with a number of optional painting settings such as color, size, effect, etc. After a short period of study, users can easily use the system to complete the VR painting experience.

Index Terms: Applied computing—Arts and humanities—Fine arts; Software and its engineering—Software creation and management—Designing software—Software design engineering

1 INTRODUCTION

The digital painting system has numerous advantages, including low cost, high flexibility, editability, and ease of saving. VR technology will bring users an immersive experience similar to painting in the real world. Furthermore, VR can overcome the limitation of screen space, allowing users to appreciate paintings from all angles and directions. The key to a successful VR painting system is to establish a highly realistic painting effect, a realistic virtual environment, and positive user interaction experiences.

Digital painting systems can realize a variety of artistic styles of painting, including watercolor, oil painting, sketches, and so on. This paper's research focuses on watercolor painting. Watercolor painting is widely employed in art creation, commercial entertainment, and other sectors as a significant form of aesthetic expression. The commonly used scheme is to use pigment simulation to realize the process of watercolor painting. This scheme produces more authentic watercolor painting works, and the user can engage in more interactive watercolor painting. Benjamin et al. [2] proposed a particle-based model to simulate watercolor pigments and complete watercolor painting with triangles as the basic element. Chu et al. [3] implemented Expressii, a real-time watercolor painting system based on the Lattice-Boltzmann Method, the system can simulate real-world watercolor paintings on hardware equipped with gravity sensors. These methods have achieved realistic watercolor effects. However, this work still has some shortcomings. On the one hand, the interaction mode of these systems is very different from the actual painting process, increasing the learning cost for users. On the other hand, these painting systems are based on fluid simulation, which will conduct discrete operations in space during fluid simulation. Such painting results can only be produced at a fixed resolution. After the painting results are enlarged, there will be an unclear problem, which makes it impossible for users to observe closely or enlarge the painting details for appreciation. In addition, adopting

a fluid simulation-based painting system to accomplish detailed watercolor painting necessitates a significant amount of calculation.

To address these problems, we implemented a VR-based watercolor painting system. To generate a realistic watercolor effect, the system employs a vector drawing method based on the Temporal Diffusion Curve (TDC) [4]. TDC is an improved method of vector graph generation based on Diffusion Curve [5]. The drawings are in the form of vector drawings. In addition to reducing the amount of calculation, it also ensures that the painting works can be kept clear under close observation. The system features natural interaction as well as a number of adjustable settings such as color, size, watercolor effects, and color mixing method. In the experiment, after a few minutes of introduction and learning, the user can draw very effective watercolor paintings.

2 DRAWING ALGORITHM

We adopt a method based on the method of Li et al. [4] to make vector artworks based on strokes. The painting's creation is divided into three steps: TDC fitting, stroke effect processing, and color computation.

To create a painting, the approach employs strokes as the primary element. And strokes are obtained by TDC fitting, each TDC corresponds to a stroke. During the painting process, the system will generate control points at fixed intervals. After obtaining the control points, the cubic spline curve interpolation method is used to fit TDC, and the cubic spline curve through all the control points is used to express TDC in the form of parametric equations. The natural boundary conditions are incorporated into the solution procedure.

After fitting a TDC curve, it is important to apply a stroke effect to the curve. The most basic and important aspect of making a stroke in realistic watercolor painting is the diffusion of watercolor pigments. Therefore, in the effect processing stage, the TDC needs to be diffused first to cause a stroke. Given that the finite difference approach is typically employed to solve the two-dimensional diffusion equation, the result obtained is a discrete solution, and the final painting result can only be shown as a raster image. So we adopt the method of Fourier transformation. Finally, to obtain the calculation result, the system just needs to calculate a one-dimensional curve integral, significantly lowering the amount of work. After obtaining the initial concentration diffusion results, the concentration results must be adjusted in order to enrich the diffusion effect of the strokes. Firstly, the unequal diffusion effect is accomplished by adding paper texture. The sigmoid, power, and Laner-Jones functions are then employed to accomplish the uniformizing, feathering, and deep edge effects, respectively.

Finally, the system calculates the color of the stroke according to the concentration and color parameters. The computer uses the RGB model for color display, but this model is not suitable for painting. Therefore, the color is first converted to RYB model before calculation and then converted back to the RGB model after calculation. The average color of each TDC is obtained by using the concentration diffusion result as the weight of the color parameter. Then, multiply the resulting transparency channel of the average

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Figure 1: Virtual scene of painting system

color by the final concentration to get the transparency of the stroke. For color mixing between brushes, a multi-layer model is adopted. According to the diffusion time of TDC, TDC can be divided into active TDC and stationary TDC. Each active TDC corresponds to one active layer in the model, and all stationary TDCs correspond to one stationary layer. In the process of color mixing, it is only necessary to mix the active layer and the static layer in turn, according to the order of drawing. Each mixing must only take into account the color mixing between the two adjacent layers. In color mixing, two methods are used: transparency color mixing and pigment color mixing.

3 PAINTING SYSTEM

The drawing system was developed in Unity and the interaction is realized based on Steam VR. As shown in Fig. 1, the whole scene is a studio. In a virtual scene, the user sits in front of an easel with paper. There are optional painting parameters on both sides of the easel. Optional parameters include size, type, and mixed type. Size stands for stroke thickness, and the system provides 5 options on a scale of 1 to 5. The type of watercolor effect of the stroke is represented by the system, and there are three options: smooth, exponent, and dark edge. Mixed type stands for stroke color blending type, the system provides two types, respectively transparency and pigment. In addition, optional parameters on a scale of 1 to 5 are provided for each type. There is a color palette for selecting colors. The color palette realizes a discrete color selector [1]. The palette offers 50 colors that are selected from the "Faber Castell Polychromos Pastel Paints"¹. This discrete color selector offers a limited variety of colors, but it will make it easier for users to choose colors and improve the user experience.

The virtual reality equipment required for the system includes a head-mounted display and a handle. The user moves the gamepad and pulls the trigger on the gamepad to complete the painting and select the painting parameters. Users can freely choose the color and other painting parameters. The strokes are displayed on the paper in real time.

4 EXPERIMENT

To test the drawing effect and practicability of the system, we invited 5 participants (N = 5; 2 females) to use the system to create paintings. All participants were able to draw using the system after about five minutes of learning. The average time it took to complete the painting was 25.3 minutes. Each participant was interviewed for about 3 minutes after the painting was completed. In the interview, all participants agreed that the system was easy to use, but would cause discomfort symptoms such as muggy and eye strain due to prolonged painting. They observed clear strokes in the drawing, with no distortion, and all agreed that virtual reality and vector drawing improved the drawing experience. Four participants believed that

¹Art-Paints.com. Faber Castell Polychromos Pastel Paints. <http://www.art-paints.com/Paints/Pastel/Faber-Castell/Faber-Castell-Polychromos.html>



Figure 2: The participant is using the system in the experiment

the functions provided by the system could meet their painting needs and create colorful effects, but another participant believed that the colors of the system were not comprehensive enough and needed to be replaced with other similar colors.

In Fig. 2, the participant wearing HMD and holding a handle is drawing using the system. In the painting as shown in Fig. 1, the participant used a variety of drawing effects, for example, the smooth effect is used to paint mountains, the dark edge effect is used to paint balloons, the exponent effect is used to paint kites, transparency color mixing is used to process the color mixing of balloons, pigment color mixing is used to process the color mixing of trees and flowers.

5 CONCLUSION

Compared to the traditional digital painting system, our system uses VR technology to provide users with an immersive painting experience. Compared to the existing VR painting system, such as Google Tilt Brush, our system realizes 2D painting that is closer to real life. Meanwhile, vector painting algorithm used by the system improves the painting experience by allowing users to see clear strokes in the virtual environment. The system is very easy to use and provides users with a variety of optional parameters, beginners can draw colorful watercolor effects after a short period of learning.

In future work, we will enrich the drawing algorithm of the system and provide users with more styles of drawing functions. It will also make the system's edits better and give users a wider range of colors to choose from. In the future, it is hoped that VR technology can be further applied in the field of art by further improving our system.

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